Exercice 1 :

1/ L'expression de H est donné par : $H = -\frac{1}{2}\sum_{ij} J_{ij} S_i S_j - \sum_i B_i S_i$

or $Z = \sum_{\{S\}} e^{-\beta H}$

$\Rightarrow Z = \sum_{\{S\}} e^{\beta/2 \sum_{ij} J_{ij} S_i S_j + \beta \sum_i B_i S_i}$

$Z = \sum_{\{S\}} \prod_{i,j} e^{\beta/2 J_{ij} S_i S_j} \prod_{i=1}^{N} e^{\beta B_i S_i}$

/ $\langle S_i \rangle = \sum_{\{S\}} S_i \frac{e^{-\beta H}}{Z}$

$= \frac{1}{Z} \sum_{\{S\}} S_i \, e^{\beta/2 \sum J_{ij} S_i S_j + \beta \sum_i B_i S_i}$

$= \frac{1}{Z} \sum_{\{S\}} \frac{1}{\beta} \times \frac{\partial}{\partial B_i} e^{\beta/2 \sum J_{ij} S_i S_j + \beta \sum_i B_i S_i}$

$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial B_i} \sum_{\{S\}} e^{\beta H}$

$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial Z}{\partial B_i} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial B_i} = K_B T \frac{\partial \ln Z}{\partial B_i}$

$= \frac{\partial (K_B T \ln Z)}{\partial B_i} = -\frac{\partial F}{\partial B_i}$

or $M = \sum \langle S_i \rangle \Rightarrow M = \sum_i -\frac{\partial F}{\partial B_i}$

/ $\langle S_i S_j \rangle = \sum_{\{S\}} S_i S_j \frac{e^{\beta H}}{Z}$

$= \frac{1}{Z} \sum_{\{S\}} S_i S_j \, e^{\beta/2 \sum_{ij} B_{ij} S_i S_j} e^{\beta \sum_i B_i S_i}$

$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \sum_{\{S\}} S_j \, e^{\beta/2 \sum_{ij} B_{ij} S_i S_j} e^{\beta \sum B_i S_i}$

$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \sum_{\{S\}} Z \times \frac{S_j e^{\beta H}}{Z} = \langle S_j \rangle$

$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} Z \langle S_j \rangle$

Scanné avec CamScanner

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \left( \frac{1}{\beta} Z \frac{\partial \ln Z}{\partial B_j} \right)$$

$$= \frac{1}{\beta^2} \frac{1}{Z} \left[ \frac{\partial Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j} + Z \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} \right]$$

$$= \frac{1}{\beta^2} \left[ \frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j} + \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} \right]$$

$$\Gamma_{ij} = \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle$$

$$= \cancel{\frac{1}{\beta^2} \frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j}} + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i B_j} \cancel{- \frac{1}{\beta^2} \frac{\partial \ln Z}{\partial B_i} \frac{\partial \ln Z}{\partial B_j}}$$

$$= \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i \partial B_j} = \frac{1}{\beta^2} \frac{\partial}{\partial B_i \partial B_j} \left( \frac{-F}{k_B T} \right)$$

$$= -\frac{1}{\beta^2} \times \beta \frac{\partial F}{\partial B_i \partial B_j} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$\Gamma_{ij} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$\frac{\partial M}{\partial B_j} = \sum_i -\frac{\partial^2 F}{\partial B_i \partial B_j} = \sum_i \frac{-\beta}{\beta} \frac{\partial^2 F}{\partial B_i \partial B_j}$$

$$= \sum_j \beta \Gamma_{ij}$$

1) $B_i = B$ champ uniforme :

$$\sum_i \Gamma_{ij} = -\frac{1}{\beta} \frac{\partial^2 F}{\partial B} = \frac{1}{\beta} \frac{\partial M}{\partial B}$$

$$\chi = \beta \sum \Gamma_{ij}$$

$$\left[ \langle \xi^2 \rangle - \langle \xi \rangle^2 \right] = \frac{1}{\beta} \left[ \left\langle \left( \sum_p S_i \right)^2 \right\rangle - \left\langle \sum_p S_i \right\rangle^2 \right]$$

$$= \beta \left[ \sum_n \langle S_i^2 \rangle - \sum \langle S_i \rangle^2 \right]$$



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$$\langle S_i^2 \rangle = \sum_{\{S\}} S_i^2 \frac{e^{\beta H}}{Z} = \frac{1}{Z} \sum S_i \frac{1}{\beta} \frac{\partial}{\partial B_i} e^{\beta H}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \times \frac{\partial}{\partial B_i} \sum \frac{S_i}{Z} \frac{e^{\beta H}}{Z}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \left( \frac{1}{Z} \times \langle S_i \rangle \right)$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial}{\partial B_i} \left( \frac{1}{Z} \times \frac{1}{\beta} \frac{\partial \ln Z}{\partial B_i} \right)$$

$$= \frac{1}{Z} \frac{1}{\beta^2} \frac{\partial Z}{\partial B_i} + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$= \frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial B_i} \right)^2 + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$\langle S_i \rangle = \frac{1}{\beta} \left( \frac{\partial \ln Z}{\partial B_i} \right)$$

$$\Rightarrow \beta \left[ \langle \varepsilon^2 \rangle - \langle \varepsilon \rangle^2 \right] = \beta \left[ \cancel{\frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial B_i} \right)^2} + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial B_i^2} - \cancel{\frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial B_i} \right)^2} \right]$$

$$= \sum \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial B_i^2}$$

$$= \sum \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial B_i \partial B_i}$$

$$\Gamma_{ij} = \sum_{ij} \langle S_i S_j \rangle - \langle S_i \times S_j \rangle$$

$$= \langle \varepsilon^2 \rangle - \langle \varepsilon \rangle^2$$

$$= \beta \left[ \sum \langle S_i^2 \rangle - \sum_i \langle S_i \rangle^2 \right]$$

$$= \beta \left[ \sum_i \langle S_i \rangle - \sum_{ij} \langle S_i \times S_j \rangle \right]$$

$$= \beta \left[ \sum_{ij} \langle S_i S_j \rangle - \sum_{ij} \langle S_i \rangle \langle S_j \rangle \right]$$

$$= \beta \sum_{ij} \Gamma_{ij}$$



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1/ $\langle S_i S_j \rangle = \sum S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j + \beta^2 \sum_{ij} J_{ij} S_i S_j}}{Z} = \langle H^2 \rangle$

$= \sum_{\{S_i\}} S_i S_j \frac{e^{\beta/2 \sum_{ij} J_{ij} S_i S_j}}{Z}$

$= \frac{2}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial J_{ij}} \sum e^{\beta H}$

$= \frac{2}{\beta} \times \frac{1}{Z} \frac{\partial Z}{\partial J_{ij}} = \frac{2}{\beta} \frac{\partial \ln Z}{\partial J_{ij}}$

remarque :

$M = \sum_i S_i \qquad \hat{H} = \hat{H}_0 - M \sum_i S_i$

1/ la fluctuation de l'aimantation

$(\Delta M)^2 = \langle (M - \langle M \rangle)^2 \rangle$

$= \langle (M^2 - 2M\langle M \rangle + \langle M \rangle^2 \rangle$

$= \langle M^2 \rangle - 2\langle M \rangle \langle M \rangle + \langle M \rangle^2$

$\boxed{(\Delta M)^2 = \langle M^2 \rangle - \langle M \rangle^2}$

$\langle M \rangle = \sum_{\{S\}} M \frac{e^{-\beta H}}{Z}$

$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta(H_0 - H \sum S_i)}$

$= \frac{1}{Z} \sum_{\{S\}} M e^{-\beta(H_0 - MH)}$

$= \frac{1}{Z} \sum_{\{S\}} \frac{1}{\beta} \frac{\partial e^{-\beta(H_0 - MH)}}{\partial H}$

$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial H} \sum_{\{S\}} e^{-\beta(H_0 - MH)}$

$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial Z}{\partial H} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \Rightarrow \langle M \rangle^2 = \frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial H} \right)^2$



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$$\langle M^2 \rangle = \sum_{\{S_i\}} M^2 \frac{e^{-\beta H}}{Z}$$

$$= \frac{1}{Z} \sum_{\{S_i\}} M^2 e^{-\beta(H_0 - HM)}$$

$$= \frac{1}{Z} \times \frac{1}{\beta} \sum_{\{S_i\}} \frac{\partial}{\partial H} \left( e^{-\beta(H_0 - HM)} M \right)$$

$$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial H} \left( \sum_{\{S_i\}} e^{-\beta(H_0 - HM)} M \right)$$

$$= \frac{1}{\beta} \times \frac{1}{Z} \frac{\partial}{\partial H} \left( Z \times \sum \frac{e^{-\beta(H_0 - HM)}}{Z} M \right)$$

$$= \frac{1}{\beta} \cdot \frac{1}{Z} \frac{\partial}{\partial H} \left( Z \times \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \right)$$

$$= \frac{1}{\beta^2} \frac{1}{Z} \frac{\partial}{\partial H} \left( Z \frac{\partial \ln Z}{\partial H} \right)$$

$$= \frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial H} \frac{\partial \ln Z}{\partial H} + \frac{\partial^2 \ln Z}{\partial H^2} \right)$$

$$= \frac{1}{\beta^2} \left( \left( \frac{\partial \ln Z}{\partial H} \right)^2 + \frac{\partial^2 \ln Z}{\partial H^2} \right)$$

$$\Rightarrow (\Delta M)^2 = \frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial H} \right)^2 + \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial H^2} - \frac{1}{\beta^2} \left( \frac{\partial \ln Z}{\partial H} \right)^2$$

$$= \frac{1}{\beta^2} \frac{\partial^2 \ln Z}{\partial H^2}$$

$$\text{or} \quad \chi_T = \frac{\partial M}{\partial H} = \frac{\partial}{\partial H} \left( \frac{1}{\beta} \frac{\partial \ln Z}{\partial H} \right) = \frac{1}{\beta} \frac{\partial^2 \ln Z}{\partial H^2} = \beta (\Delta M)^2$$

$$= \frac{1}{K_B T} (\Delta M)^2$$

$$\Rightarrow \boxed{(\Delta M)^2 = K T \chi_T}$$

**2)**

$$\sum_{ij} \Gamma_{ij} = \sum_{ij} \langle S_i \rangle \langle S_j \rangle - \langle S_i \rangle \langle S_j \rangle$$

(...) invariante par translation : $\langle S_i \rangle = \langle S_j \rangle = \langle S \rangle$

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$$\sum_j \Gamma_{ij} = \sum_j \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle = \langle M^2 \rangle - \langle M \rangle^2$$

$$\sum_j \Gamma_{ij} = \frac{\chi_T}{\beta}$$

$$(\Delta H)^2 = \langle (H - \langle H \rangle)^2 \rangle = \langle H^2 \rangle - \langle H \rangle^2$$

$$\langle H \rangle = \sum_{\{S_i\}} H \frac{e^{-\beta H}}{Z} = \frac{1}{Z} \sum_{\{S_i\}} H e^{-\beta H}$$

$$= -\frac{1}{Z} \sum_{\{S_i\}} \frac{\partial}{\partial \beta} e^{-\beta H} = -\frac{1}{Z} \frac{\partial}{\partial \beta} \sum_{\{S_i\}} e^{-\beta H}$$

$$= \frac{-1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle H^2 \rangle = \sum_{\{S_i\}} H^2 \frac{e^{-\beta H}}{Z} = -\frac{1}{Z} \sum_{\{S_i\}} \frac{\partial}{\partial \beta} (H e^{-\beta H})$$

$$= -\frac{1}{Z} \frac{\partial}{\partial \beta} \sum_{\{S_i\}} H e^{-\beta H}$$

$$= -\frac{1}{Z} \frac{\partial}{\partial \beta} \left( Z \times \underbrace{\sum_{\{S_i\}} \frac{H e^{-\beta H}}{Z}}_{\langle H \rangle} \right)$$

$$= \frac{1}{Z} \frac{\partial}{\partial \beta} \left( Z \times \frac{\partial \ln Z}{\partial \beta} \right)$$

$$= \frac{1}{Z} \left[ \frac{\partial Z}{\partial \beta} \times \frac{\partial \ln Z}{\partial \beta} + Z \times \frac{\partial^2 \ln Z}{\partial \beta^2} \right]$$

$$= \left( \frac{\partial \ln Z}{\partial \beta} \right)^2 + \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$\langle H^2 \rangle - \langle H \rangle^2 = \cancel{\left( \frac{\partial \ln Z}{\partial \beta} \right)^2} + \frac{\partial^2 \ln Z}{\partial \beta^2} - \cancel{\left( \frac{\partial \ln Z}{\partial \beta} \right)^2} = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$C_v = \frac{\partial \langle H \rangle}{\partial T} = \frac{\partial}{\partial T} \left( -\frac{\partial}{\partial \beta} \ln Z \right) = \frac{\partial \beta}{\partial T} \times \frac{\partial}{\partial \beta} \left( -\frac{\partial}{\partial \beta} \ln Z \right)$$

$$= +\frac{1}{k_B T^2} \frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$= \frac{1}{k_B T^2} (\Delta H)^2$$



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$$\Rightarrow \boxed{(\Delta H)^2 = K_B T^2 C_v}$$

4/ $\Delta H = (K_B T^2 C_v)^{1/2} = K_B^{1/2} T C_v^{1/2}$

or on sait que $\langle H \rangle = \frac{3}{2} N K_B T$

$$\Rightarrow \frac{\partial \langle H \rangle}{\partial T} = \frac{3}{2} N K_B = C_v$$

on remplace dans $(\Delta H)$ : $\Delta H = K_B^{1/2} T \left(\frac{3}{2} N K_B\right)^{1/2}$

$$= \left(\frac{3}{2}\right)^{1/2} K_B T N^{1/2}$$

$$= \left(\frac{3}{2}\right)^{1/2} \times \frac{1}{\beta} N^{1/2}$$

$$\Delta H \propto N^{\frac{1}{2}}$$

Exercice 3

1/ a/ $t \to 0$ $\quad f(t) = A t^{1/2} + B t^{1/4} + C t$

$$f(t) = t^{1/4} \left( A t^{1/2 - 1/4} + B + C t^{1 - 1/4} \right)$$

$$= t^{1/4} \left( A t^{1/4} + B + C t^{3/4} \right)$$

et on sait que : $\lambda = \lim_{t \to 0} \frac{\ln |f(t)|}{\ln |t|}$

$$\Rightarrow \ln f(t) = \frac{1}{4} \ln t + \ln \left( A t^{1/4} + B + C t^{3/4} \right)$$

$$\Rightarrow \frac{\ln f(t)}{\ln t} = \frac{1}{4} + \frac{\ln \left( A t^{1/4} + B + C t^{3/4} \right)}{\ln t}$$

$$\lim_{t \to 0} \frac{\ln f(t)}{\ln |t|} = \frac{1}{4} + \lim_{t \to 0} \frac{\ln \left( A t^{1/4} + B + C t^{3/4} \right)}{\ln |t|}$$

$$\Rightarrow \boxed{\lambda = \lim_{t \to 0} \frac{\ln f(t)}{\ln |t|} = \frac{1}{4}}$$



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b) $f(t) = At^{2/3}(t+B)^{2/3}$

$\ln f(t) = \ln\left(At^{-2/3}(t+B)^{2/3}\right)$

$= -\frac{2}{3}\ln At + \frac{2}{3}\ln(t+B)$

$= -\frac{2}{3}\ln A - \frac{2}{3}\ln t + \frac{2}{3}\ln(t+B)$

$\lim_{t\to 0} \frac{\ln f(t)}{\ln t} = \left(-\frac{2}{3}\right) - \frac{2}{3}\frac{\ln A}{\ln t} + \frac{2}{3}\frac{\ln(t+B)}{\ln t} \Rightarrow \boxed{\lambda = \frac{-2}{3}}$

/ $f(t) = At^2 e^{-t}$

$\ln f(t) = \ln(At^2 e^{-t}) = \ln A + 2\ln t + \ln e^{-t}$

$= 2\ln t + \ln A - t$

$\lambda = \lim_{t\to 0} \frac{\ln f(t)}{\ln(t)} = 2 + \frac{\ln A}{\ln t} - \frac{t}{\ln t} = 2.$

$\boxed{\lambda = 2}$

/ $f(t) = At^2 e^{1/t}$

$\ln f(t) = \ln(At^2 e^{1/t}) = \ln A + 2\ln t + \frac{1}{t}$

$\lim_{t\to 0} \frac{\ln f(t)}{\ln(t)} = 2 + \frac{\ln(Ae^{1/t})}{\ln t}$ $\Rightarrow$ indéfini

$f(t) = A\ln(\exp(1/t^4) - 1)$

$t \to 0$, $t^4 \to 0$ ; $\frac{1}{t^4} \to \infty$

$\exp\left(\frac{1}{t^4}\right) - 1 \sim \exp\left(\frac{1}{t^4}\right)$

$f(t) = A\ln\left(\exp\left(\frac{1}{t^4}\right)\right) = \frac{A}{t^4}$

$\Rightarrow \ln f(t) = \ln A - 4\ln t \Rightarrow \lim_{t\to 0} \frac{\ln f(t)}{\ln t} = -4 + \frac{\ln A}{\ln t}$



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1 a/ $f(t) = A \ln|t| + B$ en fait un changement de variable

$u = \ln t$

$\Rightarrow$

$t \to 0$

$u \to \infty$

$f(u) = Au + B$

$$\lambda = \lim_{u \to \infty} \frac{\ln(Au+B)}{u} = \lim_{u \to \infty} \frac{\ln\left(u\left(A + \frac{B}{u}\right)\right)}{u}$$

$$= \lim_{u \to \infty} \frac{\ln u}{u} + \frac{\ln\left(A + \frac{B}{u}\right)}{u} = 0$$

b/ $f(t) = A - Bt^{1/2} \Rightarrow \ln(A - Bt^{1/2}) = \ln A$

$$\Rightarrow \lambda = \lim_{t \to 0} \frac{\ln A}{\ln t} = 0$$

c/ $f(t) = 1, \quad t < 0$

$f(t) = 2, \quad t > 0$

$$\lambda = \lim_{t \to 0} \frac{\ln f(t)}{\ln t} = 0$$

$$\lambda = \lim_{t \to 0} \frac{\ln 2}{\ln t} = 0$$

d/ $f(t) = A(t^2 + B^2)^{1/2} (\ln|t|)^2$

$$\ln f(t) = \ln A + \frac{1}{2}\ln(t^2+B^2) + 2\ln(\ln t)$$

$$\lim_{t \to 0} \frac{\ln f(t)}{\ln t} = \lim_{t \to 0} \frac{\ln A}{\ln t} + \frac{\ln B}{\ln t} + \frac{2\ln(\ln t)}{\ln t}$$

$\frac{\ln A}{\ln t} \to 0$, $\frac{\ln B}{\ln t} \to 0$, $\frac{2\ln(\ln t)}{\ln t} \to 0$

en pose $u = \ln t$

$t \to 0$

$u \to \infty$

$$\Rightarrow \lim_{\substack{t \to 0 \\ u \to \infty}} \frac{2\ln u}{u} = 0 \quad \text{et} \quad \lambda = 0$$

Scanné avec CamScanner

$f(t) = A\,t\,\ln|t| + B$

$\ln f(t) = \ln(A\,t\ln|t| + B)$

$\ln t = u \Rightarrow t = e^u$

$$\lim_{t\to 0} \frac{\ln f(t)}{\ln t} = \lim_{t\to 0} \frac{\ln(A\,u\,e^u + B)}{u} = \lim_{t\to 0} \frac{\ln B}{u} = 0$$

$\lambda = 0$

Exercice 3

a/ Mq : $f(x,y) = x^3 + y^2$ ??

on sait que $f(\lambda^a x, \lambda^b y) = \lambda f(x,y)$

$f(x,y) = x^3 + y^2$ $\quad a = \frac{1}{3}$, $b = \frac{1}{2}$

$= \lambda^{3a} x^3 + \lambda^{2b} y^2$

$= \lambda(x^3 + y^2)$

$= \lambda f(x,y)$

$f(x,y) = x^3 y^2 + x^2 y^3$

$f(\underbrace{\lambda^a x}_{x}, \underbrace{\lambda^b y}_{y}) = \lambda^{3a} x^3 \lambda^{2b} y^2 + \lambda^{2a} x^2 \lambda^{3b} y^3$ $\Rightarrow$

$\lambda^{3a}\lambda^{2b} = \lambda^{2a}\lambda^{3b}$

$\lambda^{3a+2b} = \lambda^{2a+3b} \Rightarrow 3a + 2b = 2a + 3b$

$\Rightarrow a = 1$

$b = 1$

$f(x,y) = x^{1/a} \hat{f}(x^{-b/a} y) = y^{1/b} \check{f}(y^{-a/b} x)$

sait que $f(\lambda^a x, \lambda^b y) = \lambda f(x,y)$

cas particulier $\lambda^a x = 1 \Rightarrow \lambda = \frac{1}{x^{1/a}}$

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$$f(x,y) = x^{1/a} f(1, \lambda^{-1} y)$$

$$= x^{1/a} f\left(1, \frac{y}{x^{b/a}}\right)$$

$$= x^{1/a} f\left(1, y x^{-b/a}\right)$$

$$\boxed{f(x,y) = x^{1/a} \hat{f}\left(y x^{-b/a}\right)}$$

$$f(\lambda^a x, \lambda^b y) = \lambda f(x,y) \qquad \lambda^b y = 1 \Rightarrow \lambda = \frac{1}{y^{1/b}}$$

$$f(x,y) = y^{1/b} f(\lambda^a x, 1)$$

$$= y^{1/b} f\left(\frac{x}{y^{a/b}}, 1\right)$$

$$\boxed{f(x,y) = y^{1/b} \check{f}\left(x y^{-a/b}\right)}$$

3/

$$f(x,y) = x^3 + y^2$$

$$= x^3 \hat{f}(x^{-3/2} y) = y^2 \check{f}(y^{-2/3} x)$$

$$= x^3 \left(1 + \frac{y^2}{x^3}\right) = y^2 \left(1 + \frac{x^3}{y^2}\right)$$

$$\hat{f}(x^{-3/2} y) = 1 + \frac{y^2}{x^3}$$

$$\check{f}(y^{-2/3} x) = 1 + \frac{x^3}{y^2}$$

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Exercice 6 :

$$H = aM\left(t + bM^2\right)^{\theta}$$

avec $1 < \theta < 2$ ; $t = \frac{T - T_c}{T_c}$

Calcule de $\gamma$ :

$$\frac{\partial H}{\partial M} = a\left(t + bM^2\right)^{\theta} + aM\,2b\theta M\left(t + bM^2\right)^{\theta-1}$$

pour $H \to 0$ et $M \to 0$

Donc $\frac{\partial H}{\partial M} \simeq a\,t^{\theta} \Rightarrow \chi_T = \frac{\partial M}{\partial H} = \frac{1}{a} t^{-\theta} \simeq |t|^{-\theta}$

$\Rightarrow \chi_T \simeq |t|^{-\gamma} \Rightarrow$ **$\gamma = \theta$**

Calcule de $\delta$ :

$$H = aM\left(t + bM^2\right)^{\theta}$$

$$= aM\left(bM^2\left(\frac{t}{bM^2} + 1\right)\right)^{\theta}$$

$$= aM\left(bM^2\right)^{\theta}\left(1 + \frac{t}{bM^2}\right)^{\theta}$$

$$H = M^{1+2\theta}\, a b^{\theta}\left[1 + \theta\frac{t}{bM^2}\right]$$

$$H \sim M^{1+2\theta} \iff H \sim M^{\delta}$$

Donc **$\delta = 1 + 2\theta$**

Calcule de $\beta$ :

$$m \simeq (-t)^{\beta}$$

sur $H = 0$

$$\Rightarrow H = aM\left(t + bM^2\right)^{\theta} = 0$$

$$\Rightarrow t + bM^2 = 0$$

$$\Rightarrow M^2 = -\frac{t}{b} \Rightarrow M = \left(-\frac{t}{b}\right)^{1/2}$$

$$M \simeq (-t)^{\beta} \iff M = \left(-\frac{t}{b}\right)^{1/2}$$

$\Rightarrow$ **$\beta = \frac{1}{2}$**



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On prenons $\begin{cases} \gamma + 2\beta + \alpha \geq 2 & (1) \\ \alpha + \beta(1+\delta) \geq 2 & (2) \end{cases}$

$(1-2) \Rightarrow \gamma + 2\beta + \alpha - \alpha - \beta(1+\delta) \geq 0$

$\Rightarrow \gamma + \beta - \beta\delta \geq 0$ pour $\begin{cases} \beta = 1/2 \\ \gamma = 0 \\ \delta = 1 + 2\theta \end{cases}$

$\Rightarrow \theta + \frac{1}{2} - \frac{1}{2}(1+2\theta)$

$= \theta + \frac{1}{2} - \frac{1}{2} - \theta = 0$

les exposants critiques est bien vérifié.

EXERCICE 8 :

1/ on $G(P,t) = \dfrac{\psi(r, t^{(2-\alpha)/d})}{r^{d-2+\eta}} = r^{-(d-2+\eta)} \psi(r t^{(2-\alpha)/d})$ (1)

on sait que Dapres le cours : $G(r,t) = r^{-\tau} e^{-r/\xi}$

or $\xi \sim |t|^{-\nu}$

$= r^{-(d-2+\eta)} e^{-r/t^{-\nu}}$

$= r^{-(d-2+\eta)} e^{-r t^{\nu}}$ (2)

(1) = (2) $\Rightarrow \dfrac{2-\alpha}{d} = \nu \Rightarrow$ $\boxed{d\nu = 2 - \alpha}$ Relation de Josephson

2/

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Exercice 9

$$H = -J \sum_{\langle ij \rangle} S_i S_j$$

$$Z = \sum_{\{S\}} e^{-\beta H} = \sum_{\{S\}} e^{\beta J \sum_{\langle ij \rangle} S_i S_j}$$

$$= \sum_{\{S\}} \prod_{\langle ij \rangle} e^{\beta J S_i S_j} = \sum_{\{S\}} \prod_{\langle ij \rangle} e^{L S_i S_j}$$

or $e^{L S_i S_j} = \mathrm{Ch}L + S_i S_j \mathrm{Sh}L$

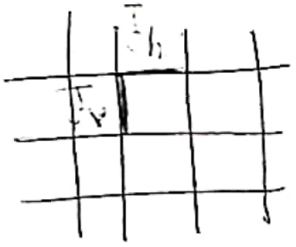
$$= (1 + S_i S_j \mathrm{th}L)\,\mathrm{Ch}L$$

$$\Rightarrow Z = \sum_{\{S\}} \prod_{\langle ij \rangle} \mathrm{Ch}L\,(1 + S_i S_j \mathrm{th}L)$$ on pose $\mathrm{th}L = u$

$$\Rightarrow Z = \sum_{\{S\}} \prod_{\langle ij \rangle} \mathrm{Ch}L\,(1 + S_i S_j u)$$

$$= (\mathrm{Ch}L)^S \sum_{\{S\}} \left[ 1 + u \sum_{ij} S_i S_j + u^2 \sum_{\langle ij \rangle \neq \langle kl \rangle} (S_i S_j)(S_k S_l) \right]$$

$$= (\mathrm{Ch}L)^S\, 2^N \left[ 1 + \sum_n \Omega_n u^n \right]$$




1) puisque on a un réseau Carré

$$H = -J_v \sum_{\langle ij \rangle} S_i S_j - J_h \sum_{\langle kl \rangle} S_k S_l$$

$$\Rightarrow Z = \sum_{\{S\}} e^{-\beta H} = \sum_{\{S\}} e^{-\beta\left(-J_v \sum_{\langle ij \rangle} S_i S_j - J_h \sum_{\langle kl \rangle} S_k S_l\right)}$$

$$= \sum_{\{S\}} e^{\beta J_v \sum_{\langle ij \rangle} S_i S_j}\, e^{\beta J_h \sum_{\langle kl \rangle} S_k S_l}$$

$$L = \beta J_v$$

$$K = \beta J_h$$

$$\Rightarrow Z = \sum_{\{S\}} e^{L \sum_{\langle ij \rangle} S_i S_j}\, e^{K \sum_{\langle kl \rangle} S_k S_l}$$

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$$\begin{cases} e^{L S_i S_j} = Ch L + S_i S_j Sh L \\ e^{K S_k S_l} = Ch K + S_k S_l Sh K \end{cases} \qquad Th K = u'$$

$$= \sum_{\{S_i\}} \prod_{\langle ij \rangle} Ch L \left(1 + S_i S_j u\right) \times \prod_{\langle kl \rangle} Ch K \left(1 + S_k S_l u'\right)$$

EXERCICE N°0:

1/ on sait que le modèle d'Ising

$$H = -J \sum_{\langle ij \rangle} S_i S_j \qquad S_i = \pm 1$$

- le modèle de Potts s'écrit comme :

$$H = -J \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j}$$

avec $\delta_{\sigma_i \sigma_j} = \dfrac{1 + \sigma_i \sigma_j}{2}$

$$\Rightarrow H = -J \sum_{\langle ij \rangle} \frac{1 + \sigma_i \sigma_j}{2}$$

$$= -\frac{J}{2} \sum_{\langle ij \rangle} \sigma_i \sigma_j + C$$

$$H = -J_{ising} \sum_{\langle ij \rangle} \sigma_i \sigma_j + C \qquad \text{pour } q = 2$$

avec $\begin{cases} \sigma_i = S_i \\ \sigma_j = S_j \end{cases} = 0$

$$\boxed{H = -J \sum_{ising \langle ij \rangle} S_i S_j}$$

* pour $q = 2 \Rightarrow \delta_{ij} = \frac{1}{2}\left[\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + 1\right]$

$$H = -J_p \sum_{ij} \delta_{ij} = -J_p \frac{1}{2} \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

sachant que :

$$\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) = \begin{cases} 1 & \text{si } n_i = n_j \\ -1 & \text{si } n_i - n_j = 1 \end{cases}$$



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$$H = -\frac{J_p}{2} \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

$$H = -J_c \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C \Rightarrow \boxed{J_c = \frac{J_p}{2}}$$

→ Potts Model est équivalent à Clock Model.

* pour $p = 3$ on pose $S_{ij} = \frac{2}{3}\left[\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + \frac{1}{2}\right]$

$$\cos\left(\frac{2\pi}{p}(n_i - n_j)\right) = \begin{cases} 1 & \text{si } n_i = n_j \\ -\frac{1}{2} & \text{si } n_i - n_j = 1 \end{cases}$$

$$H = -J_p \sum_{ij} S_{ij} = -\frac{2}{3} J_p \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

$$= -J_c \sum_{ij} \cos\left(\frac{2\pi}{p}(n_i - n_j)\right) + C$$

→ Potts Model est équivalent à Clock Model $\Rightarrow \boxed{J_c = \frac{2}{3} J_p}$



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1/ $H = -J \sum_{\langle ij \rangle} S_i S_j - H \sum_i S_i$ $\quad S_i = \pm 1$

l'énergie libre champ moyen. $F_{CM}$

$F_{CM} = \min_{\{\lambda_i\}} \{\lambda_i\}$ . $\quad H = H_0 + H_1$

$\Rightarrow H_1 = H - H_0$

-> l'inegalité de bregelbev $F \leq F_0 + \langle H - H_0 \rangle_0$

$F \leq F_0\{\lambda_i\} + \langle H\{\lambda\} - H_0\{\lambda_i\} \rangle_0 = \phi(\lambda)$

Calcule $F_0$ : $\quad F_0 = -K_B T \ln Z_0$

$H_0 = -\lambda \sum S_i$

l'champ effectif

$$Z_0 = \sum_{\{S\}} e^{-\beta H_0} = \sum_{\{S\}} e^{\beta \lambda \sum_i S_i} = \sum_{\{S\}} \prod_{i=1}^{N} e^{\beta \lambda S_i}$$

$$= \sum_{\{S\}} \prod_{i=1}^{N} \left( Ch(\beta\lambda) + S_i \, Sh(\beta\lambda) \right)$$

$$= \left( Ch(\beta\lambda) \right)^N \sum_{\{S\}} \prod_{i=1}^{N} \left( 1 + S_i \, Sh(\beta\lambda) \right)$$

$$= \left( Ch(\beta\lambda) \right)^N 2^N \Rightarrow Z_0 = \left( 2 \, Ch(\beta\lambda) \right)^N$$

donc $\boxed{F_0 = -N K_B T \ln \left( 2 \, Ch(\beta\lambda) \right)}$

Calcul de $\langle H - H_0 \rangle_0$

$$\langle H - H_0 \rangle = \langle -J \sum_{\langle ij \rangle} S_i S_j - H \sum_i S_i - \lambda \sum_i S_i \rangle$$

$$= \langle -J \sum_{ij} S_i S_j - (H + \lambda) \sum_i S_i \rangle$$

$$= -J \sum_{ij} \langle S_i S_j \rangle - (H + \lambda) \sum_i \langle S_i \rangle$$

2/ on néglige les corrélation : $\Gamma_{ij} = \langle S_i S_j \rangle - \langle S_i \rangle \langle S_j \rangle = 0$

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$\Rightarrow \langle S_i S_j \rangle = \langle S_i \rangle \langle S_j \rangle$

$\langle H - H_0 \rangle = -J \sum_{ij} \langle S_i \rangle \langle S_j \rangle - (H + \lambda) \sum_i \langle S_i \rangle$

stème est invariante par translation : $\langle S_i \rangle_0 = \langle S_j \rangle_0 = \langle S \rangle_0$

$\Rightarrow \langle H - H_0 \rangle = -J \sum_{ij} \langle S \rangle_0^2 - (H + \lambda) \sum_i \langle S_i \rangle$

$\langle H - H_0 \rangle = -J \frac{zN}{2} \langle S \rangle_0^2 - (H + \lambda) N \langle S \rangle_0$

ercices $\langle S \rangle_0$

$\langle S \rangle_0 = \langle S_i \rangle_0 = \frac{\sum_{\{S_i\}} S_i e^{-\beta H_0}}{\sum_{\{S_i\}} e^{-\beta H_0}} = th(\beta\lambda)$

$\Rightarrow \boxed{\langle S \rangle_0 = th(\beta\lambda)}$

$\Rightarrow \boxed{\langle H - H_0 \rangle = -J \frac{zN}{2} th^2(\beta\lambda) - (H + \lambda) N th(\beta\lambda)}$

$\lambda) = -N k_B T \ln\left(2 ch(\beta\lambda)\right) - \frac{J z N}{2} th^2(\beta\lambda) - (H - \lambda) N th(\beta\lambda)$

$\frac{d\lambda}{\lambda} = 0 \Rightarrow \lambda_{min} = ?$

$= -N k_B T \, 2\beta \frac{sh(\beta\lambda)}{2 ch(\beta\lambda)} - \frac{J z N}{2} \times 2 \times th(\beta\lambda) \times \left(th(\beta\lambda)\right)' + N th(\beta\lambda)$

$- (H - \lambda) N \left(th(\beta\lambda)\right)' = 0$

$-N th(\beta\lambda) - J z N th(\beta\lambda) \cdot \beta \frac{1}{ch^2(\beta\lambda)} + N th(\beta\lambda) - (H - \lambda) \frac{N\beta}{ch^2(\beta\lambda)} = 0$

$\frac{-N\beta}{ch^2(\beta\lambda)} \left\{ J z th(\beta\lambda) + (H - \lambda) \right\} = 0$

$J z th(\beta\lambda) + (H - \lambda) = 0$

$\Rightarrow \boxed{\lambda_{min} = \underbrace{J z th(\beta\lambda)_{min}}_{\langle S \rangle_0} + H}$

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aussi $\boxed{\lambda_{min} = \beta J z \langle S \rangle_0 + H}$

alors :

$$F_{CH} = -NK_B T \ln\left(2\mathrm{Ch}(\beta\lambda_{min})\right) - \frac{JzN}{2}\mathrm{th}^2(\beta\lambda_{min}) - (H - Jz\,\mathrm{th}(\beta\lambda_{min}) - H)N\,\mathrm{th}(\beta\lambda_{min})$$

$$\boxed{F_{CM} = -NK_B T \ln\left(2\mathrm{Ch}(\beta\lambda_{min})\right) + \frac{JzN}{2}\mathrm{th}^2(\beta\lambda_{min})}$$

or $\langle S \rangle_0 = \mathrm{th}(\beta\lambda_{min})$

$$= \mathrm{th}\left(\beta\left(Jz\,\mathrm{th}(\beta\lambda_{min}) + H\right)\right)$$

$$\boxed{\langle S \rangle_0 = \mathrm{th}\left(\beta\left(Jz\langle S_0 \rangle + H\right)\right)}$$ l'éq d'état

finalement :

$$\begin{cases} F_{CM} = -NK_B T \ln\left(2\mathrm{Ch}\left(\beta\left(Jz\langle S \rangle_0 + H\right)\right)\right) + \frac{JzN}{2}\mathrm{th}^2(\beta\lambda_{min}) \\ \langle S \rangle_0 = \mathrm{th}\left(\beta\left(Jz\langle S_0 \rangle + H\right)\right) \end{cases}$$

où $\langle S \rangle_0$ est la solution de l'équation d'état du modèle

$\mathrm{th}(\beta J z \langle S \rangle) = 0$ $\quad \beta J z = 1$ $\quad \Rightarrow \frac{Jz}{K_B} = T_c$

$\Rightarrow Jz = K_B T_c$

2/ $\langle S \rangle_0 = \mathrm{th}(\beta\lambda_{min}) = \mathrm{th}\left(\beta\left(Jz\langle S_0 \rangle + H\right)\right)$

si $H = 0 \Rightarrow \langle S \rangle_0 = \mathrm{th}(\beta Jz \langle S \rangle_0)$

on utilise le développement de $\mathrm{th}(x) = x - \frac{x^3}{3}$

$$\langle S \rangle_0 = \mathrm{th}(\beta Jz \langle S \rangle_0) = \beta Jz \langle S \rangle_0 - \frac{(\beta Jz \langle S \rangle_0)^3}{3}$$

$$\Rightarrow \langle S \rangle_0 = \frac{T_c}{T}\langle S \rangle_0 - \frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle S \rangle_0^3$$

$$\Rightarrow \left(1 - \frac{T_c}{T}\right)\langle S \rangle_0 = -\frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle S \rangle_0^3$$

$$\Rightarrow \langle S \rangle_0^2 = -3\left(1 - \frac{T_c}{T}\right)\left(\frac{T}{T_c}\right)^3$$

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$$\langle S\rangle_c^2 = -3\left(\frac{T}{T_c}-1\right)\left(\frac{T}{T_c}\right)^2$$

$$\langle S\rangle_0 \simeq -(t)^{1/2} \sim (-t)^{\beta} \Rightarrow \boxed{\beta = \frac{1}{2}}$$

$$\langle M\rangle = th\left(\beta(H + J\mathcal{Z}\langle S\rangle)\right)$$

$$= th\left(\beta J\mathcal{Z}\langle S\rangle + \beta H\right)$$

$$= th\left(\beta J\mathcal{Z}\left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right)\right)$$

$$= th\left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right)$$

à $T \sim T_c^-$

$J\mathcal{Z} = k_B T_c$

$\Rightarrow \beta J\mathcal{Z} = \frac{k_B T_c}{k_B T_c} = 1$

$$\langle M\rangle = \langle S\rangle = \left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right) - \frac{1}{3}\left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right)^3$$

$$\Rightarrow \frac{H}{J\mathcal{Z}} = \frac{1}{3}\left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right)^3$$

$$\Rightarrow H = \frac{J\mathcal{Z}}{3}\left(\langle S\rangle + \frac{H}{J\mathcal{Z}}\right)^3$$

$$\Rightarrow H \sim \langle S\rangle^3 \Leftrightarrow H \sim \langle M\rangle^3 \Rightarrow \boxed{\delta = 3}$$

) on sait que : $\chi_T = \frac{\partial\langle M\rangle}{\partial H} = \frac{\partial\langle S\rangle}{\partial H}$

avec $\langle S\rangle_0 = th\left(\beta(J\mathcal{Z}\langle S\rangle_0 + H)\right)$

$$\frac{\partial\langle S\rangle}{\partial H} = \left(\beta + \beta J\mathcal{Z}\frac{\partial\langle S\rangle_0}{\partial H}\right) th'\left(\beta J\mathcal{Z}\langle S\rangle_0 + \beta H\right)$$

$$th'(x) = \frac{1}{Ch^2(x)} = 1 - th^2(x)$$

$$\frac{\partial\langle S\rangle}{\partial H} = \chi_T = \left(\beta + \beta J\mathcal{Z}\chi_T\right)\left[1 - th^2\left(\beta J\mathcal{Z}\langle S\rangle_0 + \beta H\right)\right]$$

$$\chi_T = \left(\beta + \beta J\mathcal{Z}\chi_T\right)\left(1 - \langle S\rangle_0^2\right)$$

$$\chi_T = \chi_T\left(\frac{\beta}{\chi_T} + \beta J\mathcal{Z}\right)\left(1 - \langle S\rangle_0^2\right)$$

$$1 = \frac{\beta}{\chi_T}\left(1 - \langle S\rangle_0^2\right) + \beta J\mathcal{Z}\left(1 - \langle S\rangle_0^2\right)$$



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$T < T_c \qquad (S \neq 0)$

$$\chi = \frac{1 - \langle S \rangle_0^2}{J\beta(1 + \langle S \rangle_0^2)} \qquad m = \langle S \rangle \sim (-t)^{\beta}$$

donc $\beta = 1/2$

$$= \frac{1 - t}{J\beta(1+t)} \simeq \frac{1-t}{2t} = \frac{1}{2t} - \frac{1}{2}$$

$$\chi = \frac{1}{2t} \sim (t)^{-1} \Rightarrow \gamma = 1$$

6/ on sait que $\text{th}(x) \simeq x - \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

$$\langle S \rangle = \text{th}(\beta J z \langle S \rangle + \beta H) = \text{th}\left(\beta J z \left(\langle S \rangle + \frac{H}{Jz}\right)\right)$$

pour $H = 0 \Rightarrow \langle S \rangle = \text{th}(\beta J z \langle S \rangle)$

$\Rightarrow$

$$\langle S \rangle = \beta J z \langle S \rangle - \frac{1}{3}(\beta J z)^3 \langle S \rangle^3 + \frac{2}{15}(\beta J z)^5 \langle S \rangle^5$$

$$= \left(\frac{T_c}{T}\right)\langle S \rangle - \frac{1}{3}\left(\frac{T_c}{T}\right)^3 \langle S \rangle^3 + \frac{2}{15}\left(\frac{T_c}{T}\right)^5 \langle S \rangle^5 \quad \text{avec } t = \frac{T - T_c}{T_c}$$

$$= \frac{1}{t+1}\langle S \rangle - \frac{1}{3}\left(\frac{1}{t+1}\right)^3 \langle S \rangle^3 + \frac{2}{15}\left(\frac{1}{t+1}\right)^5 \langle S \rangle^5$$

$$\Rightarrow 1 - \frac{1}{t+1} = -\frac{1}{3}\left(\frac{1}{t+1}\right)^3 \langle S \rangle^2 + \frac{2}{15}\left(\frac{1}{t+1}\right)^5 \langle S \rangle^4$$

$$t \times \left(\frac{1}{t+1}\right) = -\frac{1}{3}\left(\frac{1}{t+1}\right)^3 \langle S \rangle^2 + \frac{2}{15}\left(\frac{1}{t+1}\right)^5 \langle S \rangle^4$$

$$\Rightarrow t = -\frac{1}{3}\left(\frac{1}{t+1}\right)^2 \langle S \rangle^2 + \frac{2}{15}\left(\frac{1}{t+1}\right)^4 \langle S \rangle^4$$

$$\Rightarrow \langle S \rangle^2 = \frac{t}{-\frac{1}{3}\left(\frac{1}{t+1}\right)^2} + \mathcal{E}(t^4)$$

$$\langle S \rangle^2 = -3t(1+t)^2 \Rightarrow \langle S \rangle = \sqrt{-3t}(1+2t)^{1/2}$$

$$\simeq \sqrt{-3t}\left(1 - 2 \times \frac{1}{2}t\right)$$

$$\simeq \sqrt{-3t}(1 + t^2)$$

$$\boxed{\langle S \rangle^2 \simeq -3t(1+2t)}$$

$$\boxed{\langle S \rangle \simeq \sqrt{(-3t)(1+2t)}} \Rightarrow \boxed{\beta = 1}$$

Scanné avec CamScanner

$$\frac{\beta}{\chi_T}\left(1-\langle S\rangle_0^2\right) = 1 - \beta J z\left(1-\langle S\rangle_0^2\right)$$

$$\frac{\chi_T}{\beta\left(1-\langle S\rangle_0^2\right)} = \frac{1}{1-\beta J z\left(1-\langle S\rangle_0^2\right)}$$

$$\Rightarrow \chi_T = \frac{\beta\left(1-\langle S\rangle_0^2\right)}{1-\beta J z\left(1-\langle S\rangle_0^2\right)}$$

$$= \frac{\beta\left(1-\langle S\rangle_0^2\right)}{\beta\left(\frac{1}{\beta} - J z\left(1-\langle S\rangle_0^2\right)\right)}$$

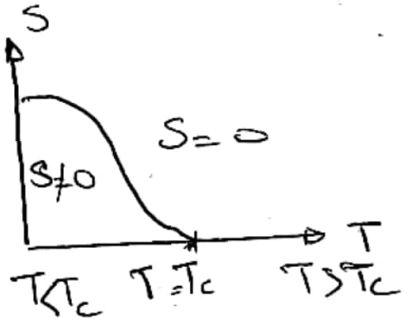
$$= \frac{1-\langle S\rangle_0^2}{k_B T - J z\left(1-\langle S\rangle_0^2\right)}$$

$$= \frac{1-\langle S\rangle_0^2}{J z\left(\frac{k_B T}{J z} - 1 + \langle S\rangle_0^2\right)}$$

$$= \frac{1-\langle S\rangle_0^2}{J z\left(\frac{T}{T_c} - 1 + \langle S\rangle_0^2\right)} = \frac{1-\langle S\rangle_0^2}{J z\left(t + \langle S\rangle_0^2\right)}$$

($t = \frac{T}{T_c} - 1$)

/ $$\chi_T = \frac{1-\langle S\rangle_0^2}{J z\left(t+\langle S\rangle_0^2\right)}$$




pour $T > T_c \rightarrow S$

$$\Rightarrow \chi_T \simeq \frac{1}{J z t} = \frac{1}{J z} t^{-1} \sim |t|^{-1}$$

or $\chi_T \sim |t|^{-\gamma} \Rightarrow \gamma = 1$

Scanné avec CamScanner

Exercice 14

1/ $F_{cm} = -NK_BT \ln\left|2\cosh(\beta\lambda_{min})\right| + \frac{\partial\beta N}{2} \hbar^2 (\beta\lambda_{min})$

$= -NK_BT \ln\left|2\cosh\left(\beta\partial\beta\langle S\rangle + \beta H\right)\right| + \frac{\partial\beta N}{2}\hbar^2(\beta\partial\beta\langle S\rangle + \beta H)$

pour $H = 0$

$\Rightarrow F_{cm} = -NK_BT \ln\left(2\cosh|\beta\partial\beta\langle S\rangle|\right) + \frac{\partial\beta N}{2}\hbar^2(\beta\partial\beta\langle S\rangle)$

$F_{cm} = -NK_BT \ln\left(2\cosh|\beta\partial\beta\langle S\rangle|\right) + \frac{\partial\beta N}{2}\langle S\rangle^2$

$F_{cm} = -NK_BT \ln 2 - NK_BT \ln\left(\cosh|\beta\partial\beta\langle S\rangle|\right) + \frac{\partial\beta N}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT \ln\left(1 + \frac{(\beta\partial\beta)^2}{2}\langle S\rangle^2\right) + \frac{\partial\beta N}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT\left(\frac{(\beta\partial\beta)^2}{2}\langle S\rangle^2 - \frac{(\beta\partial\beta)^4}{8}\langle S\rangle^4\right) + \frac{\partial\beta N}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - NK_BT\frac{(\beta\partial\beta)^2}{2}\langle S\rangle^2 - \frac{NK_BT}{8}(\beta\partial\beta)^4\langle S\rangle^4 + \frac{\partial\beta N}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - \frac{NK_BT}{2}\left(\frac{T_c}{T}\right)^2\langle S\rangle^2 + \frac{NK_BT}{8}\left(\frac{T_c}{T}\right)^4\langle S\rangle^4 + \frac{\partial\beta N}{2}\langle S\rangle^2$

$= -NK_BT \ln 2 - \frac{NK_BT}{2}\left(\frac{1}{1+t}\right)^2(3t)(1+t)^2 + \frac{NK_BT}{8}\left(\frac{1}{1+t}\right)^4(3t)^2(1+t)^4 + \frac{\partial\beta N}{2}(3t)(1+t)^2$

$= -NK_BT \ln 2 - \frac{3}{2}NK_BT(t) + \frac{NK_BT}{2}(3t)^2 + \frac{\partial\beta N}{2}(3t)(1+t)^2$

2/ . pour $T > T_c \quad S = 0$

$\Rightarrow F_{cm} = -NK_BT \ln 2 \Rightarrow C_v = -T\left(\frac{\partial^2 F}{\partial T^2}\right) = 0$

$C_v = 0$ pour $T > T_c$.

Scanné avec CamScanner

$$\mathcal{H} = -J \sum_{\langle ij \rangle} s_i s_j - H \sum_i s_i \qquad s_i = \pm 1$$

## Exercice 13

$$\mathcal{H}_0 = -\lambda \sum s_i \qquad \mathcal{H} = \mathcal{H}_0(\lambda) + \mathcal{H}_1(\lambda)$$

H ⊙⊙

$$\mathcal{H}_1(\lambda) = \mathcal{H} - \mathcal{H}_0(\lambda) = -J \sum_{\langle ij \rangle} s_i s_j - (H - \lambda) \sum_i s_i$$

$$\langle \mathcal{H}_1 \rangle_0 = \langle H - \mathcal{H}_0 \rangle_0 = -J \sum_{\langle ij \rangle} \langle s_i \rangle \langle s_j \rangle_0 - (H - \lambda) \sum_i \langle s_i \rangle_0$$

invariante par translation : $\langle s_i \rangle_0 = \langle s_j \rangle_0 = \langle s \rangle_0$

$$\Rightarrow \langle \mathcal{H}_1 \rangle_0 = -J \Big( \sum_{\langle ij \rangle} 1 \Big) \langle s \rangle_0^2 - (H - \lambda) \Big( \sum_i 1 \Big) \langle s \rangle_0$$

$$= -J \frac{zN}{2} \langle s \rangle_0^2 - (H - \lambda) N \langle s \rangle_0$$

$$F(\lambda) = F_0(\lambda) + \langle \mathcal{H}_1 \rangle_0 \cdot \qquad \langle s \rangle_0 = th(\beta\lambda)$$

$$\mathcal{Z}_0 = \left[ 2\, ch(\beta\lambda) \right]^N \qquad F_0(\lambda) = -N k_B T \ln \left[ 2 ch(\beta\lambda) \right]$$

$$F(\lambda) = \Phi(\lambda) = -N k_B T \ln \left[ 2 ch(\beta\lambda) \right] - \frac{zJN}{2} th^2(\beta\lambda) - (H - \lambda) N th(\beta\lambda)$$

$$\left. \frac{\partial F}{\partial \lambda} \right|_{\lambda = \lambda^* = \lambda_{min}} = 0$$

$$\Leftrightarrow \quad -N \cancel{th(\beta\lambda^*)} - \frac{zJN}{2} \cdot 2\beta \frac{th(\beta\lambda^*)}{ch^2(\beta\lambda^*)} + N \cancel{th(\beta\lambda^*)} - (H - \lambda^*) N \frac{\beta}{ch^2(\beta\lambda^*)} = 0$$

$$\Rightarrow \quad -\frac{N}{ch^2(\beta\lambda^*)} \left\{ \beta J z\, th(\beta\lambda^*) + \beta (H - \lambda^*) \right\} = 0$$

$$\Rightarrow \quad \beta J z\, th(\beta\lambda^*) + \beta (H - \lambda^*) = 0$$

$$\Rightarrow \quad \boxed{J z\, th(\beta\lambda^*) = \lambda^* - H} \qquad th(\beta\lambda^*) = \frac{\lambda^* - H}{Jz}$$

$$F_{CM} = \Phi(\lambda^*) = -N k_B T \ln \left[ 2 ch(\beta\lambda^*) \right] - \frac{zJN}{2} \frac{(\lambda^* - H)^2}{(Jz)^2} + N \frac{(\lambda^* - H)^2}{Jz}$$

$$= -N k_B T \ln 2 ch(\beta\lambda^*) + \frac{N}{2Jz} (\lambda^* - H)^2$$

$$m = \langle s \rangle_0 = \frac{-1}{N} \frac{\partial F_{CM}}{\partial H} = \frac{1}{Jz} (\lambda^* - H) \Rightarrow \boxed{Jz \langle s \rangle_0 + H = \lambda^*}$$

$m$ : aimantation locale

$M = N m$ ; donc :

$$\boxed{m = \langle s \rangle_0 = th \left[ \beta (Jz \langle s \rangle_0 + H) \right]}$$

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$T < T_c$ $\quad m = \text{th}(\beta J z m)$

$H = 0$

$$\sim \beta J z m - \frac{1}{3}(\beta J z)^3 m^3$$

$$\Rightarrow 1 = \beta J z - \frac{1}{3}(\beta J z)^3 m^2$$

$$\Rightarrow \frac{1}{3}(\beta J z)^3 m^2 = \beta J z - 1$$

on pose: $y = \text{th}(\beta J z m) \simeq \beta_c J z m = m$

à $T = T_c \Rightarrow \beta_c J z = 1 \Rightarrow \boxed{J z = k_B T_c}$

$$\Rightarrow m^2 \frac{1}{3}\left(\frac{\beta}{\beta_c}\right)^3 = \frac{\beta}{\beta_c} - 1 = \frac{T_c}{T} - 1 = \frac{T_c - T}{T}$$

? l'ordre le plus bas $\Rightarrow m^2 \propto T_c - T \Rightarrow m \propto (T_c - T)^{1/2}$

$$\Rightarrow \boxed{\beta = 1/2}$$

3°/ $T = T_c$ $\quad m = \text{th}(m + \beta_c H) \approx m + \beta_c H - \frac{1}{3}(m + \beta_c H)^3$

$$m = \text{th}(m + \beta_c H) \sim m + \beta_c H - \frac{1}{3}(m + \beta_c H)^3$$

$$\Rightarrow H \sim (m + \beta_c H)^3$$

$$\sim m^3 + \dots$$

$$\Rightarrow \boxed{\delta = 3}$$

4°/ $m = \text{th}\,\beta(J z m + H) = \text{th}\left(\frac{J z}{k_B T} m + \frac{H}{k_B T}\right)$

$m = \langle \sigma \rangle_0$ $\quad T = T_c \Rightarrow \beta_c J z = 1 \Rightarrow J z = k_B T_c$

$$\frac{J z}{k_B T} = \frac{T_c}{T} = \frac{1}{1+t} \qquad t = \frac{T - T_c}{T_c}$$

$$\Rightarrow m = \text{th}\left(\frac{1}{1+t} m + \frac{1}{J z} \cdot \frac{H}{1+t}\right)$$

$$m = \text{th}\left(\frac{1}{1+t} m + \frac{1}{J z} \cdot \frac{H}{1+t}\right)$$

$\text{th}(x) \simeq x - \frac{x^3}{3}$ $\quad x = \frac{1}{(1+t)} m + \frac{1}{J z} \cdot \frac{H}{(1+t)}$

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$$\frac{\partial m}{\partial H} = \frac{\partial x}{\partial H} - x^2 \frac{\partial x}{\partial H} = (1-x^2)\frac{\partial x}{\partial H}$$

avec $\quad \frac{\partial x}{\partial H} = \frac{1}{(1+t)}\left(\frac{\partial m}{\partial H} + \frac{1}{Jz}\right)$

$$\lim_{H\to 0}(1-x^2) = \frac{(1+t)^2 - m^2}{(1+t)^2}$$

$$\chi = \left.\frac{\partial m}{\partial H}\right|_{H\to 0} = \frac{((1+t)^2 - m^2)}{(1+t)^3}\left(\chi + \frac{1}{Jz}\right)$$



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donc on trouve (after few transformation) $\quad \chi = \frac{1}{Jz}\left[\frac{(1+t)^2 - m^2}{t + m^2}\right]$

$1 \gg t \Rightarrow$ $\boxed{\chi = \frac{1-m^2}{Jz(t+m^2)}}$ ou $\chi = \frac{1 - \langle s\rangle_0^2}{Jz(t+\langle s\rangle_0^2)}$

5% . à $t>0$, $m=0$ (paramagnétique)

$$\chi \simeq \frac{1}{Jzt} \sim t^{-1} \quad ; \chi \sim |t|^{-\gamma} \Rightarrow \boxed{\gamma = 1}$$

. à $t<0$ (ferromagnétique)

$$m^2 = \langle s\rangle_0^2 = -3t \Rightarrow \chi \simeq \frac{1+3t}{Jz(-2t)} \sim \frac{-1}{2Jzt} \sim \dots t^{-1}$$

$$\chi \sim |t|^{-\gamma}$$

l'exposant critique $\boxed{\gamma = 1}$

$\Rightarrow$

6% $\quad \langle s\rangle_0 = \text{th}(\beta J z \langle s\rangle_0) = \text{th}\left(\frac{1}{t+1}\langle s\rangle_0\right)$

$H=0$

$$= \frac{\langle s\rangle_0}{t+1} - \frac{\langle s\rangle_0^3}{3(1+t)^3} + O\left(\frac{\langle s\rangle_0^5}{(1+t)^5}\right)$$

$$= \langle s\rangle_0 (1-t) - \langle s\rangle_0^3/3 + O\left(\langle s\rangle_0 t^2, \langle s\rangle_0^3 t, \langle s\rangle_0^5\right)$$

$$\Rightarrow \quad -t = \langle s\rangle_0^2/3 + O\left(t^2, \langle s\rangle_0^2 t, \langle s\rangle_0^4\right)$$

$$\Rightarrow \quad -t \approx \langle s\rangle_0^2/3 \quad \Rightarrow \quad \langle s\rangle_0 \approx (-3t)^{1/2}$$

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Cont...

$$\chi = \frac{(1+t)^2 - m^2}{(1+t)^3}\,\chi + \frac{1}{\sqrt{3}}\,\frac{(1+t)^2 - m^2}{(1+t)^3}$$

$$\Rightarrow \left\{1 - \frac{(1+t)^2 - m^2}{(1+t)^3}\right\}\chi = \frac{1}{\sqrt{3}}\left(\frac{(1+t)^2 - m^2}{(1+t)^3}\right)$$

$$\Rightarrow \chi = \frac{(1+t)^3 - (1+t)^2 - m^2}{\sqrt{3}\left[(1+t)^2 - m^2\right]} \qquad \langle S\rangle_0 = -3t^{1/2}(1+\Omega t)$$

$$= \frac{1}{\sqrt{3}}\left\{\frac{(1+t)^3 - (1+t)^2 - m^2}{(1+t)^2 - m^2}\right\}$$

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$$(1+t)^3\,\chi = \chi\left[(1+t)^2 - m^2\right] + \frac{1}{\sqrt{3}}\left((1+t)^2 - m^2\right) \qquad t = -\frac{\langle S\rangle_0^2}{3}$$

$$\Rightarrow \chi = \frac{}{\sqrt{3}\left((1+t)^2 - m^2\right)}$$

$$\left((1+t)^3 - (1+t)^2 + m^2\right)\chi = \frac{(1+t)^2 - m^2}{\sqrt{3}}$$

$$\Rightarrow \chi = \frac{(1+t)^2 - m^2}{\sqrt{3}\,t\,(t+m^2)} = \frac{1 - m^2}{\sqrt{3}\,t\,(t+m^2)}$$

$$\tanh x \underset{0}{\simeq} x - \frac{x^3}{3} + \frac{2}{15}x^5 + O(x^7) \qquad (\text{"th = odd function of x"})$$

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